Customer Segmentation Using Hierarchical Clustering

Minor Project Report, AI/ML

Submitted by: [Your Name], [Roll No.], [Branch/Semester], [College Name]. Guided by: [Teacher's Name]. Date: October 2026

Abstract

Customer segmentation divides a customer base into groups of similar customers so that businesses can design targeted marketing strategies. This project applies hierarchical (agglomerative) clustering to the Mall Customers dataset, which contains 200 customers described by gender, age, annual income and spending score. The features Annual Income and Spending Score were standardised, and the number of clusters was chosen using a dendrogram and the silhouette score. Four linkage methods (Ward, complete, average and single) were compared for 2 to 8 clusters. Ward linkage with 5 clusters gave the best result, with a silhouette score of 0.554 and a Davies-Bouldin index of 0.578. The five segments were interpreted as Premium Customers, Careful Spenders, Average Customers, Impulsive Spenders and Budget Customers, and marketing actions were suggested for each. The result was also compared with K-Means clustering, which performed almost identically.

1. Introduction

Businesses today collect large amounts of data about their customers, but treating all customers in the same way wastes marketing effort. Customers differ in how much they earn, how much they spend and what they value. Customer segmentation addresses this by grouping customers so that members of a group are similar to each other and different from members of other groups.

Clustering is an unsupervised machine learning technique that discovers such groups without needing labelled data. Among clustering methods, hierarchical clustering is popular because it produces a tree of nested clusters called a dendrogram. The dendrogram lets the analyst see how clusters merge at different levels of similarity and choose a suitable number of clusters, instead of fixing that number in advance.

2. Problem Statement

A shopping mall holds basic data about its customers but does not know what types of customers it has. The problem is to group the customers into meaningful segments based on their annual income and spending behaviour using hierarchical clustering, and to describe each segment so that the mall can plan targeted marketing.

3. Objectives

1. To explore and understand the customer dataset.
2. To apply hierarchical agglomerative clustering to segment the customers.
3. To choose the number of clusters using a dendrogram and the silhouette score.
4. To compare different linkage methods and select the best one.
5. To profile each segment and suggest marketing strategies.
6. To compare the hierarchical result with K-Means clustering.

4. Background

4.1 Hierarchical clustering

Hierarchical clustering builds a hierarchy of clusters. The agglomerative (bottom-up) approach used in this project starts with every customer as a separate cluster and repeatedly merges the two closest clusters until only one cluster remains. The sequence of merges is recorded and drawn as a dendrogram. The divisive (top-down) approach works in the opposite direction but is used less often.

4.2 Linkage methods

The linkage method defines the distance between two clusters:

- **Ward:** merges the pair of clusters that causes the smallest increase in total within-cluster variance. It tends to give compact clusters of similar size.
- **Complete:** uses the largest distance between points of the two clusters.
- **Average:** uses the average distance between all pairs of points.
- **Single:** uses the smallest distance between points of the two clusters. It can produce long chains of points, known as the chaining effect.

4.3 Dendrogram

A dendrogram is a tree diagram in which the height of each join shows the distance at which two clusters were merged. Cutting the tree at a chosen height gives a set of clusters. A long vertical stretch with no merges suggests a natural place to cut.

4.4 Evaluation metrics

- **Silhouette score** ranges from -1 to 1. Higher values mean that points are close to their own cluster and far from other clusters. Values above 0.5 are generally considered good.
- **Davies-Bouldin index** measures the average similarity between each cluster and its most similar cluster. Lower values are better.
- **Calinski-Harabasz index** is the ratio of between-cluster to within-cluster dispersion. Higher values are better.

5. Dataset Description

The Mall Customers dataset (obtained from Kaggle) contains 200 records and 5 columns.

Column	Description
CustomerID	Unique ID of the customer (1 to 200)
Gender	Male or Female (about 56% female and 44% male)
Age	Age of the customer (18 to 70 years)
Annual Income (k$)	Annual income in thousands of dollars (15 to 137)
Spending Score (1-100)	Score assigned by the mall based on customer behaviour and spending (1 to 99)

The dataset has no missing values. Annual Income and Spending Score were used as clustering features because they directly describe a customer's purchasing power and purchasing behaviour.

6. Methodology

The project followed these steps:

1. **Data loading and exploration:** the data was loaded with pandas and checked for shape, missing values and summary statistics.
2. **Exploratory data analysis:** histograms of age, income and spending score, a gender count plot and a correlation heatmap were produced.
3. **Feature selection and scaling:** Annual Income and Spending Score were selected and standardised with StandardScaler so that both features contribute equally to the distance calculation.
4. **Dendrogram:** a dendrogram with Ward linkage was drawn to estimate the number of clusters.
5. **Model comparison:** agglomerative clustering was run with four linkage methods for 2 to 8 clusters, and the silhouette score was recorded for each combination.
6. **Final model:** the best configuration was used to assign every customer to a cluster, and the clusters were evaluated with three metrics.
7. **Profiling:** the average age, income and spending of each cluster were computed and each cluster was given a descriptive name.
8. **Comparison:** the result was compared with K-Means clustering using the same number of clusters.

7. Implementation

The project was implemented in Python 3.11 using a Jupyter notebook in Visual Studio Code. The following libraries were used:

Library	Purpose
pandas, numpy	Data handling and numerical operations
matplotlib, seaborn	Visualisation
scipy	Linkage matrix and dendrogram
scikit-learn	Scaling, agglomerative clustering, K-Means, PCA and evaluation metrics

8. Results and Discussion

8.1 Choosing the number of clusters

[Insert Figure 1: Dendrogram (Ward linkage). File: outputs/04_dendrogram.png]

In the dendrogram, there is a long vertical stretch with no merges between heights of roughly 5 and 9. Cutting the tree within this range gives 5 clusters, which indicates that the five groups are well separated. The three colours in the plot come from the plotting library's automatic colour threshold and do not mean three clusters.

8.2 Comparison of linkage methods

[Insert Figure 2: Silhouette score by linkage and number of clusters. File: outputs/05_linkage_comparison.png]

The silhouette score was computed for each linkage method and for 2 to 8 clusters. Ward linkage reached its highest score at 5 clusters (about 0.55) and declined afterwards. Complete linkage also peaked at 5 clusters with almost the same score. Average linkage peaked later, at 7 clusters, with a lower value than Ward at 5 clusters. Single linkage performed worst, with its best score at 2 clusters and a steady decline after that, because of the chaining effect. Ward linkage with 5 clusters was therefore selected, which agrees with the choice suggested by the dendrogram.

8.3 Final model evaluation

Metric	Hierarchical (Ward, 5 clusters)	K-Means (5 clusters)
Silhouette score (higher is better)	0.554	0.555
Davies-Bouldin index (lower is better)	0.578	0.572

A silhouette score of 0.554 indicates that the clusters are reasonably well separated.

8.4 Cluster visualisation

[Insert Figure 3: Customer segments, income versus spending score. File: outputs/06_clusters.png]

The scatter plot of income against spending score shows five visually distinct groups. A second plot using the first two principal components [Insert Figure 4: outputs/07_pca_clusters.png] shows the same five groups. Because only two features were used, PCA only rotates the axes, so this plot confirms separability rather than adding new information.

8.5 Cluster profiles

Cluster	Customers	Avg. age	Avg. income (k$)	Avg. spending score	Segment
0	32	41.0	89.4	15.6	Careful Spenders
1	39	32.7	86.5	82.1	Premium Customers (target)
2	85	42.5	55.8	49.1	Average Customers
3	21	25.3	25.1	80.0	Impulsive Spenders
4	23	45.2	26.3	20.9	Budget Customers

The largest segment is Average Customers (85 customers, 42.5% of the total), who have middle income and middle spending. Premium Customers (39 customers) combine high income with high spending and are the most valuable group. Careful Spenders (32 customers) have similarly high income but spend very little, so they represent an untapped opportunity. Impulsive Spenders (21 customers) are the youngest group, with low income but high spending. Budget Customers (23 customers) have both low income and low spending.

9. Marketing Recommendations

- **Premium Customers:** loyalty programmes, early access to new collections and exclusive offers to retain them.
- **Careful Spenders:** personalised offers and quality-focused promotions to encourage them to spend more, since they have the income to do so.
- **Impulsive Spenders:** limited-time deals, bundles and flash sales that suit their buying style, with attention to offers they can afford.
- **Average Customers:** seasonal discounts and reward points to move them towards the premium segment.
- **Budget Customers:** low-cost promotions and discount coupons, avoiding heavy marketing spend.

10. Hierarchical Clustering versus K-Means

Both algorithms produced almost identical scores (silhouette 0.554 against 0.555), which shows that the five-cluster structure in this data is strong and does not depend on the algorithm. The main advantage of hierarchical clustering here is the dendrogram, which gives a visual justification for the number of clusters without fixing it in advance. K-Means is faster on large datasets and can directly assign new customers to a cluster, whereas hierarchical clustering cannot predict the cluster of a new point without extra steps.

11. Limitations

- The dataset is small (200 customers) and is a well-known practice dataset, so results may not directly transfer to real mall data.
- Only two features were used. Age and gender were not included in the clustering.
- Hierarchical clustering is computationally expensive for very large datasets, because it needs a distance matrix between all pairs of points.
- The Spending Score is assigned by the mall, so the segments depend on how that score was defined.

12. Conclusion and Future Scope

This project successfully segmented 200 mall customers into five meaningful groups using hierarchical clustering. The dendrogram and the silhouette analysis both pointed to five clusters, and Ward linkage gave the best results. The segments were described in business terms and linked to practical marketing actions, and the results matched those of K-Means.

In future, the work can be extended by:

1. Including age and gender as additional features.
2. Using real transaction data to build RFM (Recency, Frequency, Monetary) features.
3. Testing other clustering methods such as DBSCAN or Gaussian Mixture Models.
4. Building a web application that assigns new customers to a segment in real time.

References

1. Ward, J. H. (1963). Hierarchical grouping to optimize an objective function. Journal of the American Statistical Association, 58(301), 236-244.

2. Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53-65.
3. Kaufman, L. and Rousseeuw, P. J. (1990). Finding Groups in Data: An Introduction to Cluster Analysis. Wiley.
4. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
5. Mall Customer Segmentation Data, Kaggle (dataset used in this project).